

Eric (Yu Ying) Chen

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EDUCATION

National Yang Ming Chiao Tung University

Hsinchu, Taiwan

Master of Science, Computer Science and Engineering

July 2023 – August 2025

Coursework Includes: Programming in UNIX, Operating System Capstone, Wireless Multimedia Networks, SDN

GPA: 4.23/4.30

National Tsing Hua University

Hsinchu, Taiwan

Bachelor of Science, Electrical Engineering and Computer Science

September 2019 – June 2023

University of Minnesota

Minneapolis, MN

Exchange Student, College of Science & Engineering

January 2023 – May 2023

Coursework Includes: Intro to Distributed Systems, Database Management System

GPA: 4.02/4.30

PROJECT

Operating System Capstone – Raspberry Pi 3 Kernel Development

NYCU, Spring 2025

Build a bare-metal UNIX-like OS on ARMv8-A (Raspberry Pi 3) in C, implementing core kernel subsystems.

- **Memory:** Implemented a 4-level virtual memory system with demand paging and mmap. Designed buddy and slab allocators for efficient physical memory management.
- **Multitasking:** Built threads, processes, system calls (fork, exec, kill, signal), and a round-robin scheduler.
- **Interrupts:** Handled UART and timer interrupts; implemented interrupt preemption and EL transitions.
- **Drivers:** Built UART and framebuffer drivers using ARM-GPU mailbox for hardware communication.

EXPERIENCE

Amazon

Taipei, Taiwan

Software Dev Engineer

April 2026 – Present

Tomofun Inc.

Taipei, Taiwan

MLOps Intern

June 2024 – April 2025

- Re-architected and optimized three production models for specialized AI hardware (**AWS Inferentia2**), achieving up to **80% compute cost reduction with no loss of accuracy**. (Blog post: tech.arui.dev/posts/inf2compiling)
- **Built and evaluated LLM inference pipelines with vLLM**, focusing on performance profiling, latency analysis, and scalable deployment.
- Engineered a serverless **MLOps pipeline for monitoring production model performance**, including data sampling, Label Studio project management, and weekly report generation using **AWS Lambda** and **CloudWatch**.

Iscom Online International Information Inc.

Taichung, Taiwan

Full-Stack Engineer

June 2023 – May 2024

- Built a management system for a company that is about 100 people.
- Used Vue.js and Python. Integrated with several Microsoft services such as Calendar, to-do list, and One Drive.

National Yang Ming Chiao Tung University

Hsinchu, Taiwan

Full-Stack Engineer

June 2023 – December 2024

- Helped maintain school systems.
- Used Vue.js and PHP for one system and Golang for the other system.
- Acquired the ability to effectively communicate and mentor team members.

Mei-Chu Hackathon

Hsinchu, Taiwan

Development Team Leader

December 2020 – November 2021

- **Led a team of 6 junior developers** and successfully launched two websites for participants to register and access event details. (Website link: 2021.meichuhackathon.org)
- Designed a **full-stack solution deployed on AWS**, using EC2, S3, and **Docker** for a scalable and reliable service.

RESEARCH

National Yang Ming Chiao Tung University

Hsinchu, Taiwan

Master's Thesis: Dockerized Medical Data Processing System

July 2023 – August 2025

- Developed a scalable **Go backend** for ingestion and orchestration of large medical imaging datasets.

- Containerized services with **Docker** to enable modularity, fault isolation, and reproducible deployments.

National Tsing Hua University

Hsinchu, Taiwan

Undergraduate Researcher

May 2022 – June 2023

- Conducted research under Professor Chun-Yi Lee on applying **Deep Reinforcement Learning** to train self-driving cars in a virtual environment.
- Publication: Vision based Virtual Guidance for Navigation (arxiv.org/abs/2303.02731v1)

SKILLS

- **Programming Languages:** C/C++, Python, Go (Golang), JavaScript, SQL
- **Technologies:** Linux, Operating Systems, Docker, AWS (EC2, S3, Inferentia), ML, Git, GitHub Actions
- **Languages:** Native in Chinese; Professional Working Proficiency in English (TOEFL: 93)
- **Awards:** *1st Runner Up* – TSMC x Microsoft Careerhack 2022; *2nd Place* – NTHU Senior Project Demo 2022